

A PROJECT REPORT

MUSIC RECOMMENDATION BASED ON USER'S FACIAL EMOTION

Submitted by

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*Project Report submitted to Department of Computer Science, Government
College Kasaragod in partial fulfillment of the requirements for the award of
Degree of BSc Computer Science.*



**GOVERNMENT COLLEGE KASARAGOD
(Affiliated to KANNUR UNIVERSITY)
Vidyanagar , Kasaragod
671123(2022-2025)**

MARCH 2025

DECLARATION

*I hereby declare that the project work titled
**“MUSIC RECOMMENDATION BASED ON USER’S
FACIAL EMOTION”** written and submitted by me is my original work. I also declare that this report has not been submitted to any other Universities or Institutions for the award of any fellowship, degree or diploma..*

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CERTIFICATE

This is to certify that the project “MUSIC RECOMMENDATION BASED ON USER’S FACIAL EMOTION” submitted in partial fulfillment of the requirement of the degree of BSc Computer Science is a result of bonafide work carried out by ” DEVIKA KARTHIK”, during the academic year 2024-2025.

Internal Project Guide

Head of the Department

External Examiner

- 1.
- 2.

Acknowledgement

I am happy to submit my project "MUSIC RECOMMENDATION BASED ON USER'S FACIAL EMOTION", where I try to build a relationship between music and emotion. I take this opportunity to express my sincere gratitude to the people who have supported me in this endeavor.

I wish to express my heartfelt thanks to our principal, **Dr. V. S. Anil Kumar**, our Head of Department, **Mr. Kumareshan**, Assistant Professor of the Department of Computer Science, and my project guide, **Ms. Shafna Majeed**, Assistant Professor of the Department of Computer Science and I also thank **Ms. Geena**, Assistant Professor of the Department of Computer Science, for her support. Mainly, I would like to express my heartfelt thanks to **Ms. Akhila** for her unwavering support, invaluable guidance, and constant encouragement throughout the project, which played a crucial role in its successful completion.

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Sincerely ,

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Abstract

"Music touches us emotionally, where words alone cannot."

My project, **MUSIC RECOMMENDATION SYSTEM BASED ON USER'S FACIAL EMOTION**, detects users' facial emotions and automatically plays music corresponding to their mood. The system analyzes real-time facial expressions through a camera and classifies emotions such as happiness, sadness, anger, surprise, neutrality, and Ahegao. Based on the detected emotion, it selects and plays music that aligns with the user's current mood.

The system utilizes a Convolutional Neural Network (CNN) to analyze facial expressions and classify emotions accurately. Songs are mapped to specific emotions, ensuring that the recommended music resonates with the user's feelings. We trained the model using 50 epochs and achieved an accuracy of 65%, demonstrating its capability in emotion detection.

However, many existing techniques rely on previous data to suggest music, making them less dynamic in responding to real-time emotions. Additionally, traditional algorithms are often slow and less accurate. Our approach enhances both speed and precision, offering a more intuitive and emotionally engaging music experience by focusing on real-time facial expression analysis rather than historical preferences.

Abbreviations

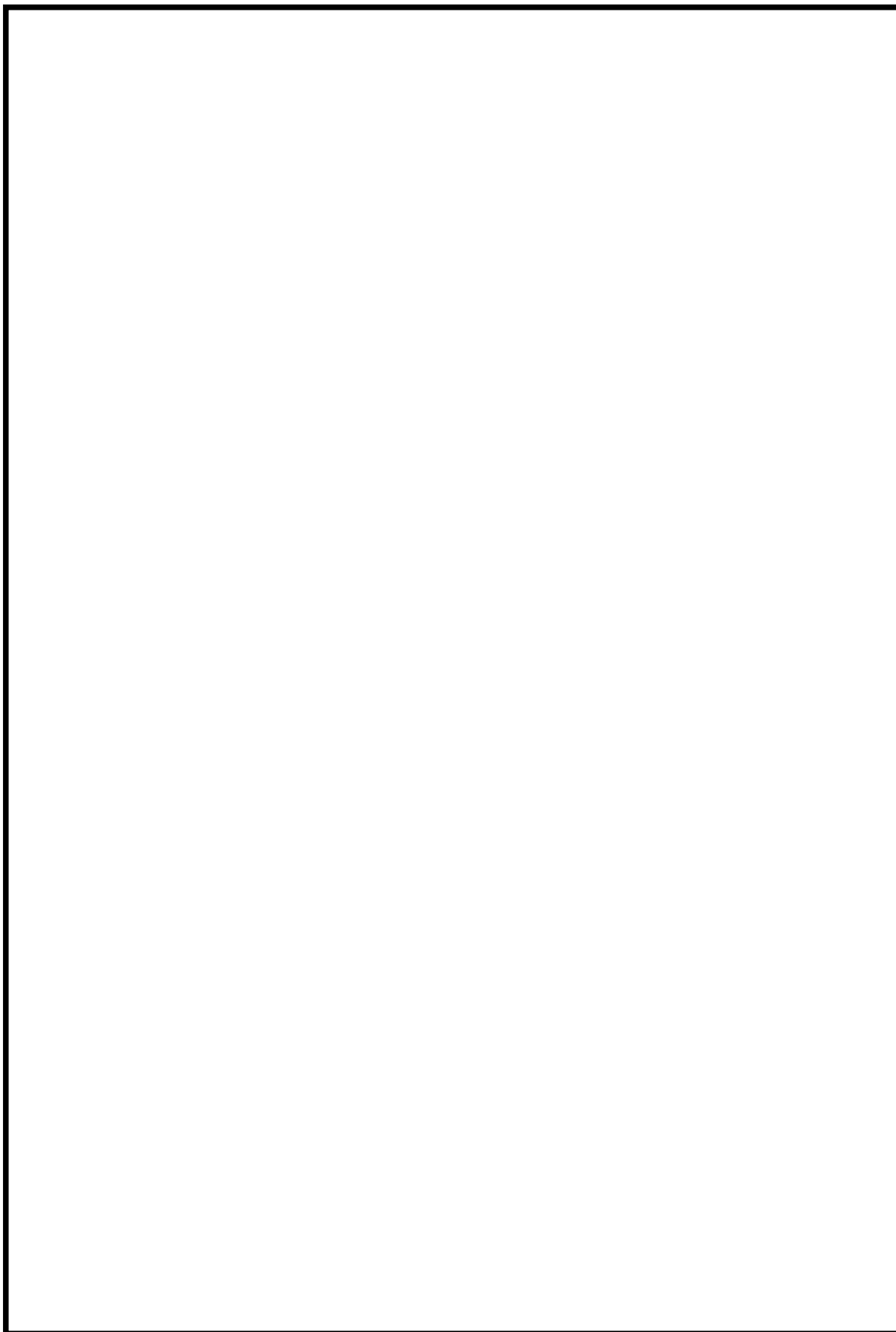
CNN	– Convolutional Neural Network
GPU	– Graphics Processing Unit
RAM	– Random Access Memory
L2	– L2 Regularization (Ridge Regularization)
ReLU	– Rectified Linear Unit (Activation Function)
Adam	– Adaptive Moment Estimation (Optimizer)
OpenCV	– Open Source Computer Vision Library
FER	– Facial Expression Recognition
TP	– True Positive (Confusion Matrix)
FP	– False Positive (Confusion Matrix)
TN	– True Negative (Confusion Matrix)
FN	– False Negative (Confusion Matrix)

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Chapter 1

Introduction

1.1 Overview

Music has the ability to evoke powerful emotional responses. It serves as an easy and effective way to change moods, relieve stress, and enhance well-being. Different forms of music reflect various emotions, and people often choose songs based on their changing moods. However, selecting music manually from a playlist can be time-consuming, and in some cases, users may resort to playing random songs that may not align with their current emotional state, leading to dissatisfaction.

Our project aims to eliminate the hassle of manual song selection by introducing an intelligent music recommendation system based on facial emotion detection. This system not only saves time but also enhances user experience by playing music that matches their current mood, thereby reducing stress, uplifting their spirits ultimately making them happy.

1.2 objective

The objective of this project is to develop an intelligent music recommendation system that detects a user's facial expression in real time and plays music

that matches their current mood. By utilizing Convolutional Neural Networks (CNNs) for emotion recognition, the system eliminates the need for manual song selection, making the process faster, more accurate, and emotionally engaging.

Key Goals:

- Enhance user experience by playing songs that align with their emotions.
- Reduce stress and uplift mood through personalized music recommendations
- Eliminate the hassle of manual song selection, saving time and effort
- Improve accuracy and efficiency using deep learning-based facial emotion detection
- Create a seamless integration between human emotions and an automated music system

1.3 Problem statement

The project aims to develop an AI-driven music recommendation system that automatically detects facial expressions and plays music that aligns with the user's emotions. By leveraging Convolutional Neural Networks (CNNs) for facial emotion recognition, the system ensures an accurate, real-time, and personalized music experience, reducing the effort needed for song selection while enhancing user satisfaction and regulating the user's mood.

1.4 Project Description

Goal: The idea is to create a fun and personalized music experience by recommending songs based on how you're feeling, all by analyzing your facial expressions. The system will detect your emotions through your webcam and play music that matches your mood.

Key Features:**Facial Emotion Recognition:**

1. Using a webcam, the system will analyze your facial expressions in real time to determine emotions like happiness, sadness, anger, surprise, or neutral.
2. We'll train a deep learning model (like a CNN) or use pre-trained models to identify these emotions and capture the subtle details of your face.

Music Recommendations:

Once the system identifies your emotion, it will suggest music that fits.

Happy: Uplifting, energetic tracks.

Sad: Relaxing or introspective songs.

Angry: Fast-paced, intense music.

Surprised: Something a little unexpected or quirky.

It'll use the tempo, genre, and other features of the music to match the vibe of your mood.

Data Sources:

We use the Facial Emotion Recognition Image Dataset from Kaggle, which contains approximately 15,000 RGB images of cropped faces labeled with corresponding emotions. A Convolutional Neural Network (CNN) is trained on this dataset to classify six emotions: happy, sad, ahogado, angry, neutral, and surprise. To enhance the user experience, each detected emotion is mapped to a set of six pre-selected songs stored in a designated folder. This approach ensures a personalized and real-time music recommendation system that aligns with the user's emotional state.

Tech Stack:

1. Python for coding and implementation.
2. OpenCV to capture video from the webcam and process facial expressions.
3. TensorFlow to train the emotion recognition model.

4. Gradio for the app's interface.

5. Audio in IPython to play music mapped to the detected emotion.

1.5 Motivation

Music has a profound impact on human emotions, often serving as a source of comfort, motivation, and relaxation. However, selecting the right music based on one's mood can sometimes be challenging and time-consuming. A music recommendation system based on facial expressions eliminates this effort by automatically suggesting songs that align with the user's current emotional state.

With advancements in artificial intelligence, computer vision, and deep learning, it is now possible to accurately detect emotions from facial expressions in real-time. This technology can be leveraged to create a more personalized and immersive music experience. By integrating emotion detection with music recommendation, users can effortlessly enjoy music that resonates with their feelings, enhancing their overall listening experience.

Furthermore, this system has significant applications in mental health and well-being. Music therapy is widely recognized for its ability to reduce stress, improve mood, and provide emotional support. A system that intuitively selects music based on a user's emotions can be beneficial for individuals struggling with anxiety, depression, or stress, offering them a seamless and therapeutic way to manage their emotions through music.

Additionally, in today's digital age, personalization is a key factor in user engagement. AI-driven solutions that adapt to human emotions are becoming increasingly relevant in various domains, from entertainment to healthcare. This project aims to bridge the gap between emotion recognition and music personalization, creating a smart, interactive, and emotion-aware recommendation system that enhances user experience and well-being.

Chapter 2

Existing System

Table 2.1: Comparison of Existing Music Recommendation Systems Based on Facial Emotion

Author	Dataset	Model	Work	Accuracy	Limitation
[1]	FER2013	CNN	Captures facial expressions using a webcam and selects songs based on emotions.	82%	Affected by lighting conditions and facial obstructions. Limited to predefined emotions.
[2]	JAFPE	CNN	Uses facial feature extraction (eyes, cheeks, lips) to classify emotions and suggest music accordingly.	85%	Can only recognize a limited number of emotions. Doesn't consider user preferences.

Author	Dataset	Model	Work	Accuracy	Limitation
[3]	CK+	CNN + Streamlit	Integrates face recognition with real-time emotion detection for personalized music recommendations.	88%	Requires high computational power for real-time processing. Privacy concerns with continuous face capture.
[4]	FER2013	CNN + SVM	Extracts facial landmarks to classify emotions and recommend music.	85%	Limited song database and emotion classification categories.
[5]	JAFPE	Deep Learning (VGG16)	Uses pre-trained deep learning model for facial emotion recognition and music suggestion.	90%	High computational cost; not suitable for low-end devices.
[6]	CK+, FER	CNN + OpenCV	Detects facial expressions in real-time and suggests songs accordingly.	83%	Struggles with mixed emotions and subtle facial expressions.
[7]	FER2013	Hybrid CNN-RNN	Combines CNN for feature extraction and RNN for sequential emotion recognition in videos.	87%	Requires a large dataset for training and validation.

Author	Dataset	Model	Work	Accuracy	Limitation
[8]	Custom Dataset	Deep CNN	Uses a custom dataset for face emotion detection and music recommendation.	80%	Poor generalization; needs a larger and more diverse dataset.
[9]	JAFFE	MLP (Multilayer Perceptron)	Uses a simple neural network for emotion recognition and music mapping.	78%	Not suitable for real-time processing; slow inference speed.
[10]	CK+	LSTM + CNN	Uses LSTM for sequence modeling and CNN for feature extraction to enhance recommendations.	89%	Complexity increases with real-time applications, leading to latency issues.

Chapter 3

Proposed System

The proposed system aims to create an interactive emotion-based music recommendation system that enhances the user's mood by playing songs based on real-time facial emotion detection.

It follows four key steps:

- 1.**Face Capture**-User capture image using webcam.
- 2.**Face Detection** – identifying facial landmarks and features using Haar cascade.
- 3.**Emotion Detection** – detecting expression and classifying into one of six emotions: happy, sad, ahogado, angry, neutral, and surprise using a trained CNN model.
- 4.**Song Playing** – Mapping the recognized emotion to a corresponding song and playing it.

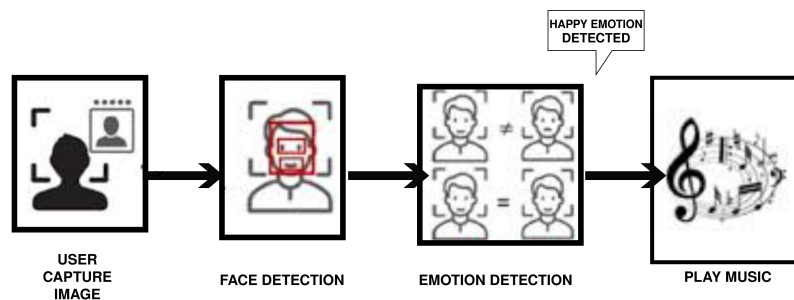


Figure 3.1: proposed system

The figure 3.1 illustrates a flowchart depicting a system that plays music based on facial expression recognition. The process begins with capturing a user's image, followed by face detection. Subsequently, the system identifies the user's emotion, specifically looking for a "happy" expression. If happiness is detected, the system proceeds to play music.

3.1 System Architecture

The system architecture of the proposed emotion-based music recommendation system consists of multiple components working together to detect emotions and recommend music accordingly. The process begins with image capture using OpenCV, which extracts facial features from webcam input. These images are then fed into a deep learning model built using TensorFlow and trained on the Facial Emotion Recognition Image Dataset from Kaggle, containing approximately 15,000 RGB images of cropped faces labeled with corresponding emotions. The model employs a Convolutional Neural Network (CNN) with multiple layers, including convolutional, pooling, and dense layers, along with a dropout layer to prevent overfitting. Since the dataset contained an unequal number of images for each emotion, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustments were applied to balance the classes.

To enhance classification accuracy, the system uses the ArcLoss function, which introduces an angular margin loss to improve feature discrimination. The final layer normalizes feature vectors, ensuring robust emotion classification into six categories: happy, sad, angry, neutral, surprise, and ahegao. Once an emotion is detected, the system maps it to a corresponding set of six pre-stored songs, stored in a designated folder, ensuring an emotion-aligned music experience. The music is played using the Audio module in IPython.

For the user interface, Gradio is utilized, providing an interactive platform where users can engage with the system seamlessly. Additionally, a random selection mode is available, allowing users to play songs unrelated to detected emotions. This structured approach ensures efficient emotion recognition and

an enhanced user experience, making the system a valuable tool for mood-based music recommendation. The figure 3.2 depicts flow diagram of the system.

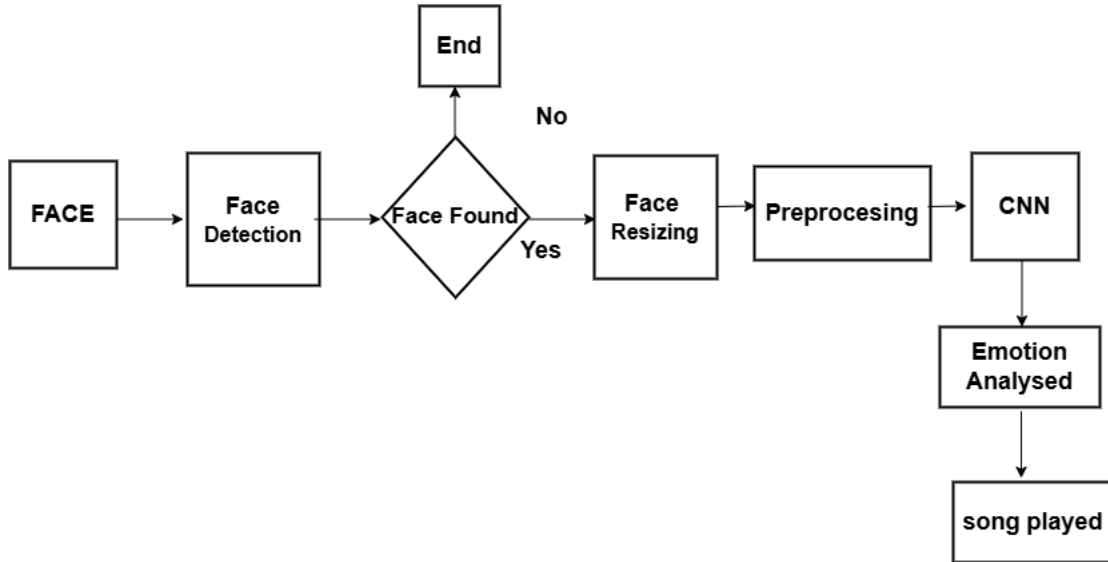


Figure 3.2: System Architecture

3.2 Methodology

Face capturing:-

The primary objective of this system is to capture images using a commonly available device, such as a webcam or other physiological devices, for real-time emotion detection. To achieve this, we utilize the OpenCV library, which integrates seamlessly with NumPy and other machine learning frameworks, making it a powerful tool for real-time computer vision applications. When the execution starts, the system accesses the camera stream and captures the image.

Preprocessing:

Face detection using OpenCV. Image normalization and resizing for deep learning model compatibility.

Face detection:-

Haar Cascade is a machine learning-based approach used for face detection. It works by training a classifier with a set of positive images (faces) and negative images (non-faces). The system scans an image at different scales to detect facial features like eyes, nose, and mouth using edge and texture patterns. OpenCV provides a pre-trained Haar Cascade model that efficiently detects faces in real time. This method is fast and lightweight, making it suitable for applications like emotion recognition and automated music recommendation based on facial expressions.

Emotion detection:-

Emotion detection is a crucial aspect of human-computer interaction, allowing systems to recognize and respond to users' emotional states. This project utilizes a Convolutional Neural Network (CNN) to classify facial expressions into six emotions: happy, sad, angry, neutral, surprise, and awe.

The system is built using TensorFlow and Keras, employing a deep learning model with multiple convolutional layers. The architecture consists of:

Convolutional Layers: Extract spatial features from facial images.

Activation function:- ReLU • Converts negative values into zero

MaxPooling Layers: Reduce dimensionality while retaining key features.

Fully Connected Layer: Processes high-level feature representations.

Dropout Layer: Prevents overfitting by randomly disabling neurons during training.

L2 Normalization: Ensures stable feature scaling for improved classification accuracy.

Loss Function – ArcLoss

This model implements ArcLoss, an additive angular margin loss function. ArcLoss enhances feature discrimination by enforcing a margin between different

emotion classes, improving classification accuracy.

Training and Emotion classification

The model is trained on the Facial Emotion Recognition Image Dataset from Kaggle, containing 15,000 RGB images categorized into six distinct emotions. Since the dataset had an unequal distribution of images per emotion, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustments were applied to balance the classes, improving generalization and overall performance. After training, it is fine-tuned using the ArcLoss function to enhance robustness and improve classification accuracy. The final layer generates a normalized vector representing the predicted emotion.

When an image passes through a Convolutional Neural Network (CNN), it undergoes a series of transformations where meaningful features are extracted. The figure 3.3 given below is the step-by-step depiction of the feature extraction process in your CNN model.

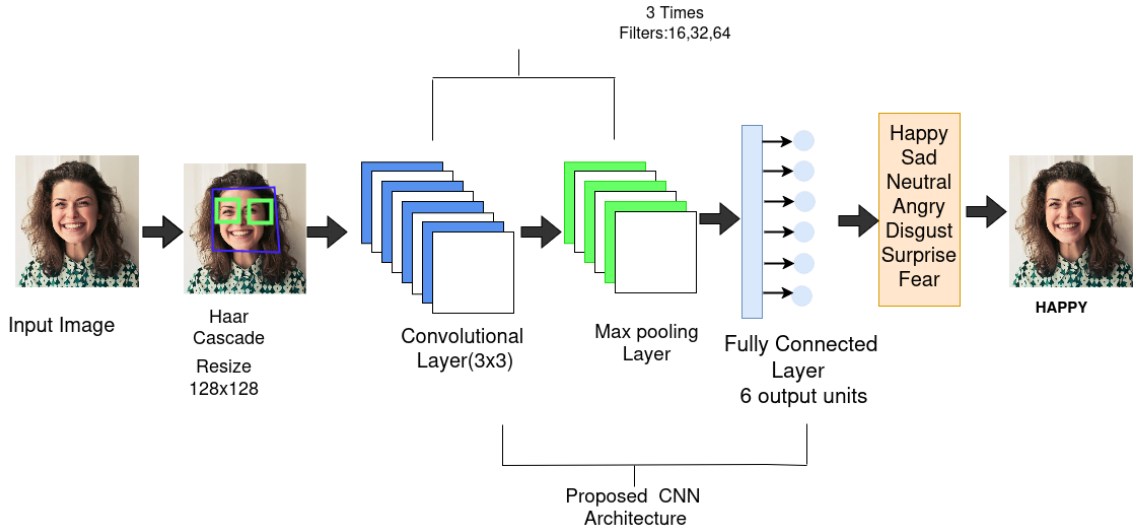


Figure 3.3: image flow through CNN

Play Music:-

Each detected emotion is mapped to a corresponding music genre and the library Audio in IPython is used to map.

Predefined emotion-music mapping database ensures consistency.

Example Mappings:

Happy → Pop, Dance, Upbeat

Sad → Soft, Acoustic, Piano

Angry → Rock, Metal, Heavy Beat

Neutral → Jazz, Lo-Fi, Classical

3.3 Experimental Setup

Dataset Used: The training dataset, sourced from **Kaggle**, consists of approximately **15,000 images** representing various facial expressions. Since the dataset had an imbalanced distribution of emotions, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustments were applied to ensure class balance, improving model generalization and performance. The emotions categorized in the dataset include **Angry, Ahegao, Sad, Happy, Neutral, and Surprise**. Before training, the images are preprocessed by resizing them to **128x128 pixels** and normalizing pixel values for optimal input representation.

Model Architecture: The model is a **Convolutional Neural Network (CNN)** with the following layers:

1. **Conv2D (16 filters, 3x3 kernel, ReLU activation)** - Extracts low-level features.

Activation Function: ReLU (Rectified Linear Unit) Applies non-linearity, helping the network learn complex patterns. Formula: $f(x) = \max(0, x)$ Ensures positive activations only.

2. **MaxPooling2D (2x2 pool size)** - Reduces spatial dimensions.

3. **Conv2D (32 filters, 3x3 kernel, ReLU activation)** - Learns more complex patterns.

4. **MaxPooling2D (2x2 pool size)** - Further reduces dimensions.
5. **Conv2D (64 filters, 3x3 kernel, ReLU activation)** - Captures high-level patterns.
6. **Flatten Layer** - Converts feature maps into a 1D vector.
7. **Dense Layer (128 neurons, ReLU activation)** - Learns deeper relationships.
8. **Dropout (50%)** - Prevents overfitting by randomly disabling neurons.
9. **Dense Layer (6 neurons, linear activation)** - Outputs a feature vector for classification.
10. **L2 Normalization** - Ensures that embeddings are normalized for distance-based classification.

Model Summary:

Layer (type)	Output Shape	Param #
conv2d_42 (Conv2D)	(None, 126, 126, 16)	448
max_pooling2d_28 (MaxPooling2D)	(None, 63, 63, 16)	0
conv2d_43 (Conv2D)	(None, 61, 61, 32)	4,640
max_pooling2d_29 (MaxPooling2D)	(None, 30, 30, 32)	0
conv2d_44 (Conv2D)	(None, 28, 28, 64)	18,496
flatten_14 (Flatten)	(None, 50176)	0
dense_28 (Dense)	(None, 128)	6,422,656
dropout_14 (Dropout)	(None, 128)	0
dense_29 (Dense)	(None, 6)	774
lambda_14 (Lambda)	(None, 6)	0

Total params: 6,447,014 (24.59 MB)
Trainable params: 6,447,014 (24.59 MB)
Non-trainable params: 0 (0.00 B)

Figure 3.4: model summary

The figure 3.4 shows the architecture of a Convolutional Neural Network (CNN) used for facial emotion recognition. It presents the model's layers, output shapes, and parameter counts. The network consists of multiple Conv2D layers for feature extraction, MaxPooling2D layers for downsampling, a Flatten layer to convert features into a 1D array, and Dense layers for classification. A

Dropout layer helps prevent overfitting, while a Lambda layer performs custom operations. The model has approximately 6.4 million trainable parameters and uses around 24.59 MB of memory. Its structured design enables effective emotion recognition from facial images.

Loss Function: The model uses **ArcLoss**, a modified softmax loss that improves class separability by enforcing angular margin constraints. It enhances the model's ability to distinguish different emotions by ensuring embeddings of different classes are further apart in feature space.

Mathematically, ArcLoss modifies the cosine similarity with a margin:

$$\cos(\theta + m) = \cos(\theta) \cdot \cos(m) - \sin(\theta) \cdot \sin(m)$$

where m is the margin and θ is the angle between feature vectors.

Optimizer: The model is trained using the **Adam optimizer** with a learning rate of **1e-4**. Adam is chosen for its adaptive learning rate and efficient convergence properties.

Model Training and Evaluation:

- **Training Size:** 80% of the dataset (10,000 images).
- **Validation Size:** 20% of the dataset (2400 images).
- **Batch Size:** 4
- **Epochs:** 50
- **Data Augmentation:** Applied techniques like rotation, zoom, flipping to improve generalization.

Evaluation Metrics: The model's performance is measured using: - **Precision:** Measures the proportion of correctly predicted positive samples out of all predicted positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** Measures the proportion of correctly predicted positive samples out

of actual positive samples.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Accuracy:** Measures overall correctness of predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

TP-total positive

TN-total negative

Fp-false positive

FN-false negative

- **Confusion Matrix:** Evaluates classification performance per class

Experimental Setup Summary

Aspect	Details
Dataset	Facial Emotion Recognition Image Dataset From Kaggle
Model	CNN-based with 3 Conv layers, 2 MaxPooling layers, Dense, Dropout, and L2 Normalization
Loss function	ArcFace Loss function (Additive Angular Margin Loss)
Optimizer	Adam optimizer (Learning rate = 1e-4)
Batch Size	4
Epochs	50
Evaluation Metrics	Precision, Recall, Accuracy

Table 3.1: Experimental Setup Summary

3.4 Hardware Requirements:

RAM: Minimum 16GB

GPU: NVIDIA GTX 1080 Ti

Storage: Adequate storage

Webcam: Required for real-time emotion detection.

3.5 Software Requirements:

Programming Language: Python 3.7 or higher

Libraries Frameworks:

TensorFlow / Keras: Used for building and training the deep learning model.

OpenCV: Used for image processing, face detection, and handling webcam input.

NumPy: Used for numerical computations and handling arrays.

Matplotlib: Used for visualizing training metrics and debugging image processing steps.

Scikit-learn: Provides utilities for model evaluation, including precision, recall, and confusion matrix calculations.

Gradio: Used to create a user-friendly web interface for real-time emotion detection and music recommendation.

Development Environment: Google Colab

Chapter 4

Performance Analysis

Training accuracy=77.49%

Training loss=8.29%

Testing accuracy=65%

Testing loss=19.17%

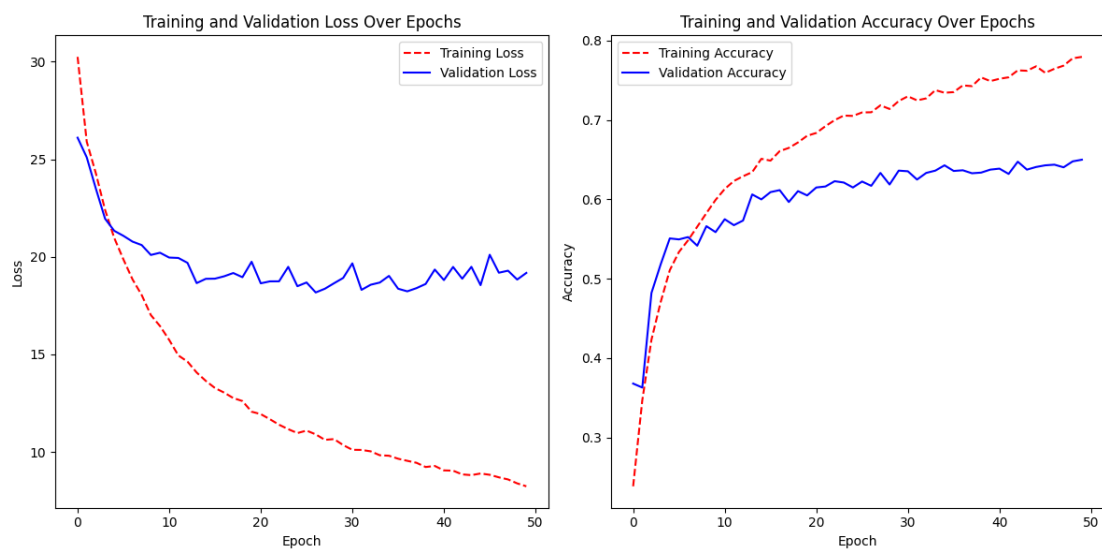


Figure 4.1: Result analysis

The graph 5.3 below shows the training and validation loss (left plot) and ac-

curacy (right plot) over epochs. In the loss graph, the training loss (red dashed line) decreases significantly at the start, showing that the model is learning. However, after around 30 epochs, the decline slows, and fluctuations appear, indicating possible overfitting. The validation loss (blue line) starts higher but steadily decreases. However, after 50 epochs, it begins to increase, suggesting that the model struggles with unseen data.

In the accuracy graph, the training accuracy (red dashed line) increases rapidly, reaching around 0.7 by epoch 30. The validation accuracy (blue line) starts lower but gradually improves, stabilizing at around 0.65. This shows that while the model performs well on training data, it struggles to generalize. The increasing validation loss and the gap between training and validation accuracy indicate overfitting. To improve generalization, techniques such as early stopping, regularization, or data augmentation can be used.

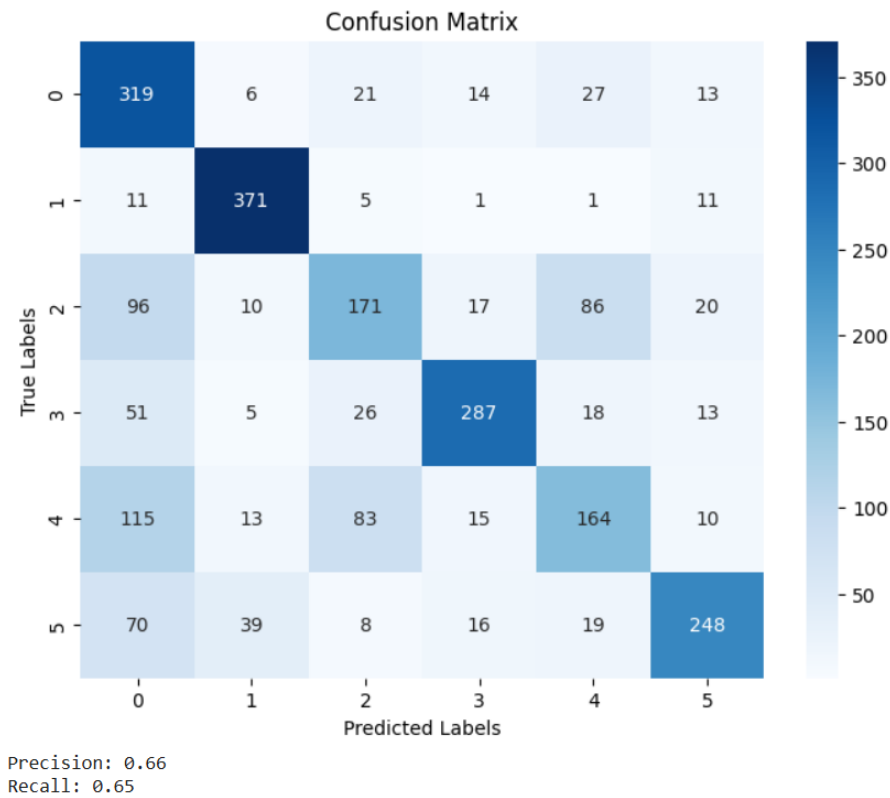


Figure 4.2: Confusion Matrix analysis

4.1 Confusion Matrix Analysis

A confusion matrix is a tool used to evaluate the performance of a classification model by comparing predicted labels with the actual label.

Understanding the Matrix

- The **rows** represent the actual (true) labels.
- The **columns** represent the predicted labels.
- The **diagonal values**(e.g,319,371,171,etc.)represent correct classifications.
- The **off-diagonal values**represent misclassified samples.

Performance Matrix

From the confusion matrix, we derive

- **Precision:0.66**-This means that when the model predicts a particular class,it is correct **66%** of the time.
- The **Rcall:0.65**-This model successfully identifies **65%** of actual cases from the dataset.

Key Observations

1. High accuracy for some classes
 - Class 0(319 correct,few misclassification) and Class 1(371 correct ,very few misclassification)have strong prediction.
2. Significant misclassification in Other classes:
 - Class 2(171 correct),but often confused with Class 0 and Class 4.
 - Class 4(164 correct),but many misclassified as Class 0 and Class 2.
 - Class 5(248 correct),but some confusion with Class 0 and Class 1
3. Potential class imbalance:

- some classes have fewer training samples to lower recall for those classes.
- Classes 0 and 1 have strong performance, while Class 2 and 4 show more confusion.

4.2 Over all Analysis

The performance of the proposed emotion-based music recommendation system was evaluated using accuracy, loss metrics, and confusion matrix analysis. Due to limited memory space, we used a restricted number of images for training, which affected the learning process. As a result, the model was slightly slow in learning and achieved an accuracy of 65%. However, we observed that with a larger dataset, the accuracy could potentially improve, as the model would have more diverse facial expressions to learn from.

The accuracy graph showed a steady increase in the early epochs, with the training accuracy improving rapidly. However, the validation accuracy plateaued around epoch 30, indicating a gap between training and validation performance. The loss analysis revealed that while the training loss decreased consistently, the validation loss started increasing after epoch 50, suggesting overfitting. This means the model performed well on the training data but struggled to generalize to new, unseen data.

The confusion matrix analysis showed that emotions like Happy and Neutral were classified with higher accuracy, whereas Angry and Surprise had more misclassifications. This suggests that some emotions have more distinct features, while others overlap, making classification more challenging. Additionally, Sad was sometimes misclassified as Neutral or Angry, while Ahegao, being more exaggerated, was often confused with Happy or Surprise due to similar facial expressions. The limited dataset size may have contributed to these misclassifications despite data augmentation efforts. Increasing the dataset size could help improve accuracy and reduce misclassifications.

In conclusion, while the model successfully detects emotions from facial expressions, challenges like overfitting and classification imbalance remain. Increasing the dataset size, enhancing augmentation techniques, and fine-tuning the network architecture could significantly boost accuracy and improve generalization.

Chapter 5

Implementation

1. Click "Generate" – The system activates the camera.
2. Capture Live Image – The user takes a photo using the webcam.
3. Preprocess the Image – The system resizes and normalizes the image.
4. Pass to Model: The processed face image is fed into the pre-trained deep learning model.
5. Detect Emotion – The model analyzes the image and predicts the emotion.
6. Play Music – A song corresponding to the detected emotion is played.

User Interface

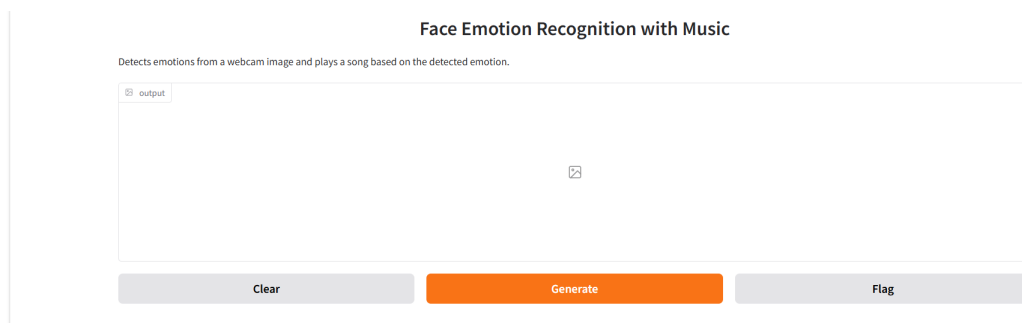


Figure 5.1: User Interface

The figure 5.3 depicts a user interface for a Face Emotion Recognition with Music system. It includes a title, a brief description, and an output section where detected emotions are displayed. There are three buttons: Generate (to capture an image and detect emotions),

Clear (to reset data), and Flag (to report issues). The layout is simple and user-friendly, allowing users to easily interact with the system.

Happy Face Detected

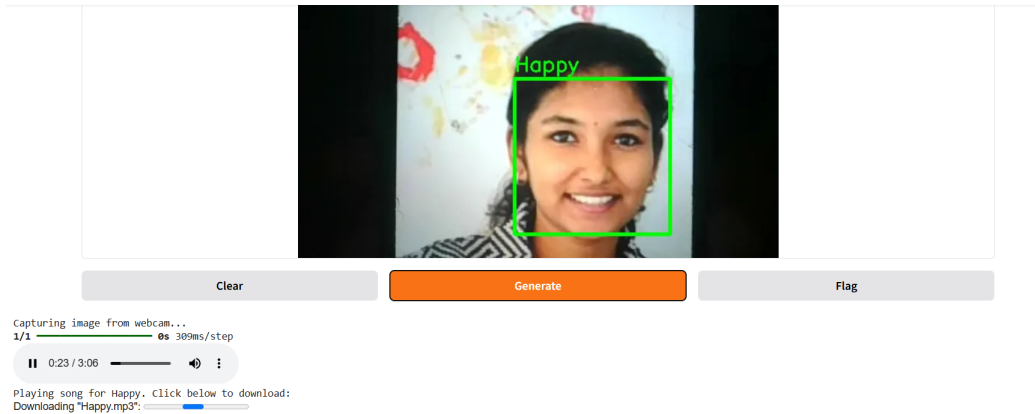


Figure 5.2: Detect Happy

The image 5.3 shows a facial recognition interface detecting happy expression. A green box highlights face, and a music player plays a song based on the emotion.

Sad Face Detected

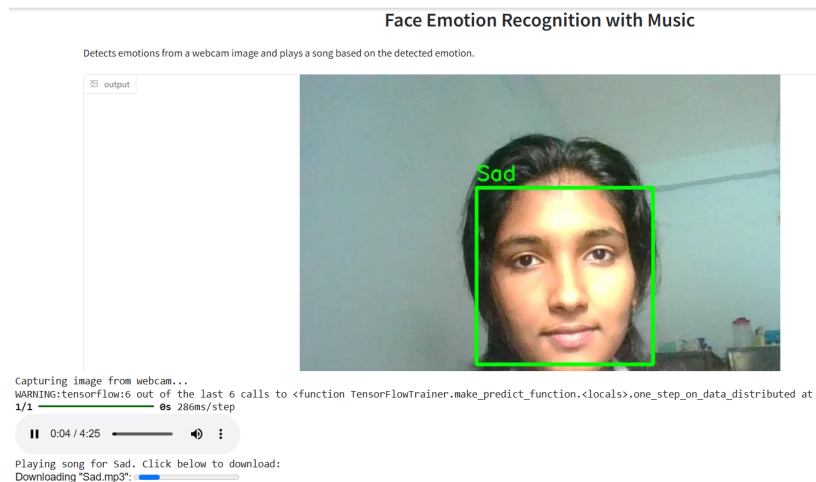


Figure 5.3: Detect Sad

The image 5.3 shows a facial recognition interface detecting sad expression.

Chapter 6

Conclusion

In conclusion, our Music Recommendation System Based on User's Facial Emotion offers a novel approach to personalized music selection by leveraging real-time facial expression analysis. By utilizing a CNN for emotion classification and mapping songs to detected moods, our system ensures a dynamic and emotionally resonant music experience.

With an achieved accuracy of 65% over 50 training epochs, our model demonstrates its effectiveness in real-time emotion detection. However, the limited dataset size due to memory constraints has impacted model performance, making learning slower. We anticipate that increasing the dataset size will further enhance accuracy.

Compared to traditional methods that rely on historical data and slower algorithms, our approach enhances responsiveness and accuracy, making music recommendations more intuitive and immersive. Future improvements, such as optimizing the model architecture and expanding the dataset, could further refine the system's performance and user experience.

Chapter 7

Future Work

The current music recommendation system is designed for local use, but future enhancements could make it more portable and efficient. Instead of relying solely on facial expressions captured by a webcam, wearable sensors can be integrated to improve accuracy.

1. Integration of Wearable Sensors:

GSR (Galvanic Skin Response): Measures skin conductivity changes, which correlate with emotional states.

PPG (Plethysmography Physiological Sensors): Tracks heart rate variations to detect stress, relaxation, or excitement.

These biometric signals, combined with facial recognition, can enhance emotion detection beyond manual image-based classification.

2. Advanced Emotion Detection:

Expanding the current system to detect additional emotions beyond the six currently used.

Using deep learning techniques to capture subtle emo-

tional changes.

3.Cloud-Based Music Recommendation:

Instead of storing songs locally, the system can be integrated with online streaming services like Spotify or Apple Music, allowing personalized music recommendations in real time.

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